9.10. A sample counted for 15 min gave 9000 total counts. A 30 min background measurement registered 1200 counts. Calculate (a) the count rate or the sample alone, with its standard deviation, and (b) with its probable error.

9.11. A certain sample has a true average counting rate of 100 cpm. What is the probability that 80 counts would be obtained in a 1 min recording?

9.14. LITERATURE

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CHAPTER 10

Energetics of Nuclear Reactions

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Reactions between an atomic nucleus and another particle are called nuclear reactions. In some such reactions, the nucleus is unchanged (*elastic scattering*), in others the original nucleus is excited to a higher energy state (*inelastic scattering*); in a third class new nuclei are formed (*nuclear transmutations*). Spontaneous nuclear transformations, which are involved in the radioactive decay of unstable nuclei, have been discussed in Chapter 5. In this chapter the emphasis is on the mass and energy relationships when a projectile interacts with a nucleus.

10.1. CONSERVATION LAWS IN NUCLEAR REACTIONS

We have previously discussed a number of laws governing nuclear processes. A summary of these laws provides us with a picture of the mechanics of nuclear interactions occurring below ~100 MeV/A. At higher energies the mass number may not be conserved.

Let us begin by considering the simple process shown in Figure 10.1, in which a *projectile*, indicated by subscript 1, collides with a *target* atom, subscript 2, forming an intermediate system, subscript i. The intermediate system splits into the products, which are designated by subscripts 3 and 4. We call this a *nuclear reaction*. The velocity of the projectile v_1 is greater than zero, but the velocity of the target atom v_2 is made to be zero by using the target nucleus as the origin of the system of reference.

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00010-0

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